

Ordinary differential equations

Project topics

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1. Design and implementation of central difference formula based discretisation and implementation of the problem

$$y''(x) + a(x)y'(x) + b(x)y(x) = f(x)$$

for all $a \leq x \leq b$ and $y(a) = \alpha$, $y(b) = \beta$. Find out the rate of convergence with respect the max-norm error. The preferred software is Matlab and the code should be interactive.

2. Chapter 2 (Will the valve hold?) (see Mathematical modelling: A case studies approach, Reinhard Illner)
3. Brachistochrone problem (Problem 32, Chapter 2, Ordinary differential equations, Boyce and DiPrima).
4. Existence and uniqueness theorem for first order ordinary differential equations using fixed point method.
5. Newton's law of gravitation and the motion of the planets (Chapter 3, Unit 21, Differential equations with applications and historical notes, G Simmons)